

Ehsan Ghasemi Firozabadi

Mechatronics Engineer · Industrial Automation · Textile Machinery Specialist

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PROFESSIONAL SUMMARY

From the start of my undergraduate studies, I joined the university robotics team and earned multiple awards in robotics competitions during my B.Sc. Based on those achievements, I was admitted to a Master's program without an entrance exam. After graduation I entered industry, worked for several years across different companies and sectors, then founded my own company, where I now serve as CEO. I have designed and built a wide range of machines for various industries and proposed many ideas to improve performance. I am a mechatronics engineer skilled in software and hardware programming, mechanical design, and electronic PCB design, with a primary passion for ideation and building new machines.

WORK EXPERIENCE

CEO — Madiar Mechatronic Company, Yazd 2016 – Present

- Lead mechatronics projects across electronics design, hardware/software programming, and machine integration.
- Deliver textile-machinery upgrades and industrial automation systems for manufacturing clients.

Electronics Designer / Hardware & Mechanics Programmer — Novin Ara Pardazesh Isatis, Yazd mid 2013 – 2016

- Designed electronic boards and programmed controllers for custom industrial machines.
- Supported mechanical design (SolidWorks) and pneumatic subsystems through commissioning.

Electronics & Electrical Designer — Ghaemiyeh Factory (Taft), Yazd 2012 – mid 2013

- Designed electrical/electronic solutions for factory production equipment.

TEXTILE / WEAVING PROJECTS

- Built a 4-color weft selector for weaving machines, USB-programmable and installable on all weaving machines
- Built a weft selector for Jacquard weaving machines
- Built waste/scrap scissors for weaving machines
- Built a dedicated encoder for Dornier weaving machines
- Built an electronic SMS control board
- Built an oil-pump control board for weaving machines
- Removed Sulzer trigger-plate sensors and replaced their function with an encoder
- Replaced Sulzer takeoff gearbox with an AC motor controlled by an inverter
- Removed packing/density gearbox on Sulzer machines and installed a servo motor instead
- Built an IoT logger for weaving machines and a mobile app for monitoring and display

OTHER PROJECTS

- Online iron-furnace temperature monitoring — Yazd Alloy Steel Company
- Smart-building motherboard — Yazd University
- Smart irrigation across 50 hectares — Yazd Alloy Steel lands

- Vital-parameter and event monitoring device for Raja trains
- Alarm-panel devices for national power substations
- Tile resistance testing machine — Yazd tile factories
- Desktop caliper device for tile factories
- Team member — natural stone sorting machine with AI
- Manufacturing of wood, stone, and plasma CNC machines

TECHNICAL SKILLS

Automation	PLC (DELTA, Fatek), HMI, WinProLadder, Fvdesigner, communication protocols
Motion	Servo, Inverter, Step Motor, pneumatics
Electronics	PCB design, Altium
MCU	STM32, STM32CubeMX, CodeVisionAVR
Mechanical	SolidWorks, industrial machine design
Software	Microsoft Office, PowerMill2018

EDUCATION

M.Sc. Mechatronics Engineering 2010 – 2013
Islamic Azad University, Qazvin

B.Sc. Electronics Engineering 2005 – 2009
Islamic Azad University, Yazd

LANGUAGES

Persian	Native
English	B1 / Advanced

PERSONAL

Date of birth	04 Aug 1987
Nationality	Iranian
Marital status	Married
Military service	Completed
Driving licence	B, C1

PATENTS

- Object localization using triangle similarity
- Mine-detection robot with mapping capability

AWARDS (ROBOTICS COMPETITIONS)

- 5th place — Singapore RoboCup — Rescue Robot — 2010
- 1st place — 3rd Kharazmi — Best Dynamics, Rescue Robot — 2010
- 2nd place — 3rd Kharazmi — Rescue Robot — 2010
- 1st place — IranOpen 2010 — Manual Mine-Detection Robot
- Nominee — Outstanding Student, IAU Yazd — 2010
- 2nd place — 2nd Kharazmi — Manual Mine-Detection Robot — 2009
- 2nd place (provincial) — Line-Follower Robot — 2009
- 3rd place — IranOpen 2009 — Manual Mine-Detection Robot
- 2nd place — 2nd Amirkabir Robotics — Painter Robot — 2009